NATHANIEL POWELL

613-222-8075 | n2powell@uwaterloo.ca | LinkedIn | github.com/Tophacks | Portfolio

EDUCATION

University of Waterloo Waterloo, ON

Bachelor of Science, Honours Mathematical Physics, Co-op Sep. 2025 - Present

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Web & Data: React, Vite, Node.js, Express, Next.js, FastAPI, PostgreSQL, Supabase, REST APIs, WebSockets
AI & Simulation: PyTorch, Stable-Baselines3, PPO, OpenCV, SUMO, TraCI, HOG/SVM
Engineering: Docker, Git, GitHub Actions, CI/CD, API testing, authentication, data modeling, Figma

EXPERIENCE

Quadmates.ca 2026 - Present

Founder & Full-Stack Developer

- Shipped a React/Express and PostgreSQL/Supabase student platform with real-time messaging, partner matching and study tracking; adapted the existing application into a Capacitor iOS beta.
- Improved chat reliability using live-event reconciliation and atomic PostgreSQL read receipts to prevent concurrent-write races; reported a 70% reduction in messaging latency (internal estimate).
- Delivered 56 successful Vercel production deployments (Oct-Nov 2026); diagnosed failed preview builds and shipped fixes to mobile design, analytics and application workflows.
- Built privacy-first analytics with anonymous sessions and Do Not Track controls, plus opt-in AI tools with validated inputs and server-side usage quotas.

Infosys InStep Jan - Apr 2027

Incoming Computer Engineering Assistant - AI Agent Engineering

- Selected for Infosys's CTO office in Bengaluru, India, for an upcoming placement focused on AI agents and enterprise process automation.

UW Orbital 2026 - Present

Business & Finance Team Member

- Negotiated corporate manufacturing and resource support for CubeSat engineering, resolving sponsor production-schedule constraints through technical documentation and compromise.
- Coordinated technical stakeholders and supported engineering budgets using Excel and Python automation prototypes.

SELECTED ENGINEERING PROJECTS

TransitOps | *Python, SUMO/TraCI, FastAPI, Next.js, Docker*

- Built a deterministic transit-control simulator modeling three buses and three signalized intersections, streaming SUMO telemetry to a web dashboard and validating integration with CI smoke tests.

NeuroSim 2.0 | *Python, PyTorch, PPO, Stable-Baselines3*

- Developed a 2D reinforcement-learning rocket-landing environment and evaluated policies by landing success, touchdown error, crashes, and fuel consumption.

Computer Vision Security Camera | *Python, OpenCV, Tkinter*

- Created real-time person detection using HOG/SVM, a multithreaded producer-consumer pipeline, bounded buffering and frame sampling.

Staffflow | *React, Authorization, Vercel CI/CD*

- Shipped staff scheduling and recruiting with role-based access and applicant workflows; Vercel recorded 47 successful production deployments (Oct-Nov 2026).